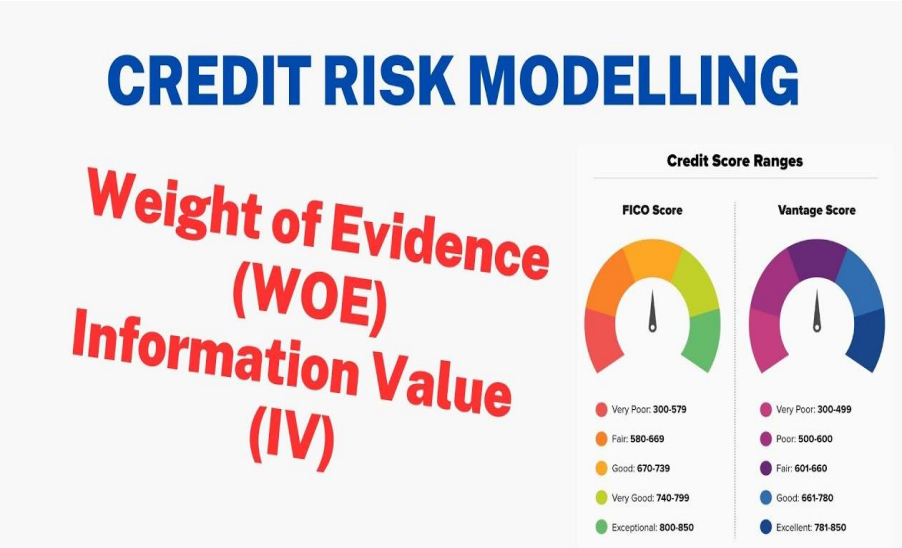


WOE & IV

ABSTRACT-

Weight of Evidence (WOE) quantifies the predictive power of a variable by measuring the log ratio of good to bad outcomes. Information Value (IV) evaluates a variable's strength in predicting a target variable. Higher IV values indicate stronger predictive power, aiding in feature selection. WOE is widely used in credit risk modeling and logistic regression. IV values help determine whether a variable should be included in a model.

Shibasish Kar

1.INTRODUCTION:-

Logistic regression model is one of the most used statistical techniques for solving binary classification problem. It is an acceptable technique in almost all the domains. These two concepts - weight of evidence (WOE) and information value (IV) evolved from the same logistic regression technique. These two terms have been in existence in credit scoring world for more than 4-5 decades. They have been used as a benchmark to screen variables in the credit risk modelling projects such as probability of default. They help to explore data and screen variables. It is also used in marketing analytics project such as customer attrition model, campaign response model etc.

2. WEIGHT OF EVIDENCE:-

The Weight of Evidence (WOE) tells the predictive power of an independent variable in relation to the dependent variable. Since it evolved from credit scoring world, it is generally described as a measure of the separation of good and bad customers. "Bad Customers" refers to the customers who defaulted on a loan. and "Good Customers" refers to the customers who paid back loan.

Weight of Evidence (WOE) is crucial in credit risk management as it helps in data transformation and predictive modelling. It ensures a monotonic relationship between independent variables and default probability, improving model interpretability. WOE minimizes noise and handles missing values effectively, making models more robust. It aids in binning variables to capture risk patterns efficiently. Additionally, WOE enhances logistic regression models, leading to better credit scoring and risk assessment.

2.1. Equal Width Binning (Fixed-Width Binning):-

- The range of the variable is divided into bins of equal size (width).
- The number of bins is predefined.
- Bin edges are determined by dividing the range of values into equal intervals.

Advantages -

- Simple and easy to implement.
- Suitable for continuous variables with a uniform distribution.

Disadvantages -

- May result in bins with very few or too many observations.
- If data is skewed, bins may not be meaningful.

2.2. Equal Frequency Binning (Quantile Binning)

The data is divided into bins so that each bin contains an approximately equal number of observations. Bin edges are chosen based on quantiles

Advantages -

- Ensures balanced bins with a similar number of observations.
- Works well with skewed data.

Disadvantages -

- Bin widths are uneven, making interpretation harder.
- Similar values might be placed in different bins.

2.3. Decision Tree Binning:-

- A decision tree algorithm (e.g., CART) is used to find the best split points in the data.
- Splits are chosen to maximize the separation between good and bad classes.

Advantages -

- Data-driven and automatic.
- Finds optimal binning for predictive power.

Disadvantages -

- May overfit without regularization.
- Needs hyperparameter tuning.

2.4. Chi-square Binning (Chi Merge):-

- Adjacent bins are merged if their distribution of good vs. bad (or other categorical outcomes) is statistically similar.
- Uses the Chi-square test to determine if two bins are significantly different.

Advantages -

- Data-driven and statistically sound.
- Ensures bins are meaningful in terms of target variable separation.

Disadvantages -

- Requires setting a significance level (e.g., 0.05).
- Computationally intensive.

2.5. Manual Binning (Domain Knowledge Binning):-

- The bins are created manually, based on expert knowledge or business rules.
- Typically used when there is prior knowledge about meaningful categories.

Advantages -

- Highly interpretable.
- Ensures bins are relevant to the business problem.

Disadvantages -

- Subjective and may introduce bias.
- May not capture data-driven patterns.

2.6. Monotonic Binning (WOE-Based Binning):-

- Ensures that WOE values increase or decrease monotonically across bins.
- If initial binning results in a non-monotonic WOE trend, bins are merged or adjusted.

Advantages -

- Ensures strong predictive power.
- Helps avoid overfitting in credit models.

Disadvantages -

- Requires manual intervention.
- May need multiple iterations.

2.7. K-means Binning:-

- Uses K-means clustering to group similar values into bins.
- The number of clusters (K) is predefined.
- Similar values are assigned to the same bin based on the clustering algorithm.

Advantages -

- Captures nonlinear relationships.
- More adaptive to data distribution.

Disadvantages -

- Computationally expensive.
- Harder to interpret compared to other methods.

2.8. Choosing the Right Binning Method:-

Method	Best For	Challenges
Equal Width	Simple data, normal distribution	Imbalanced bins
Equal Frequency	Skewed data, equal observation bins	Unequal bin widths
Decision Tree	Automated, predictive binning	Overfitting risk
Chi-square (Chi Merge)	Statistically optimized bins	Requires tuning
Manual (Expert-driven)	Business-driven binning	Subjective, may miss patterns
Monotonic Binning	Credit risk, scorecards	Needs manual adjustments
K-means Clustering	Complex distributions	Computationally heavy

To calculate Weight of Evidence (WOE) -

- Bin the Variable – Divide the independent variable into discrete bins (for categorical or continuous data).
- Calculate Good and Bad Proportions – Determine the percentage of good (non-default) and bad (default) cases in each bin

$$P(Good) = \frac{Good\ count\ in\ bin}{Total\ good\ count}$$

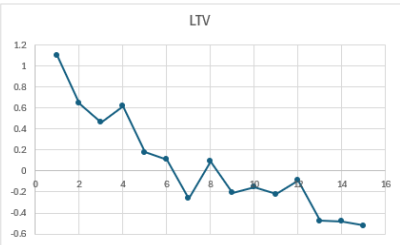
$$P(Bad) = \frac{Bad\ count\ in\ bin}{Total\ bad\ count}$$

- Compute WOE for Each Bin –

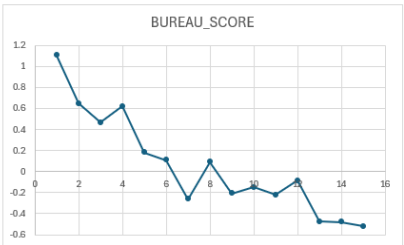
$$WOE\ = ln\left(\frac{P(Good)}{P(Bad)}\right)$$

- Handle Edge Cases – If any bin has zero bads or goods, apply smoothing (e.g., add a small constant).
- To prevent division by zero, we apply **smoothing** by adding a small constant (ϵ\epsilonϵ, e.g., 0.5 or 1) to both the good and bad counts before calculating percentages.
- Use WOE Values in Modelling – Replace original variable values with their corresponding WOE values for logistic regression or scorecard models.

LTV	BINS	TOTAL	GOODS	BADS	BADS/TOTAL	WOE	GOOD%	BAD%	WOE	IV
-Infinity	x < 72	393	362	31	0.07888041	-1.1017	0.07800	0.0259	1.1017098	0.05737782
72	72 ≤ x < 83	421	371	50	0.11876485	-0.6482	0.07994	0.0418	0.6482319	0.02471945
83	83 ≤ x < 88	373	321	52	0.13941019	-0.4643	0.06917	0.0435	0.4642502	0.0119256
88	88 ≤ x < 91	328	288	40	0.12195122	-0.6181	0.06206	0.0334	0.6181338	0.01768529
91	91 ≤ x < 94	401	330	71	0.17705736	-0.1805	0.07111	0.0594	0.1804656	0.00211881
94	94 ≤ x < 97	405	329	76	0.18765432	-0.1094	0.07089	0.0635	0.1093772	0.00080335
97	97 ≤ x < 99	319	239	80	0.2507837	0.26151	0.05150	0.0669	-0.26151	0.0040252
99	99 ≤ x < 101	457	370	87	0.19037199	-0.0916	0.07972	0.0727	0.0916477	0.00063986
101	101 ≤ x < 103	357	271	86	0.24089636	0.20818	0.05839	0.0719	-0.208176	0.00281324
103	103 ≤ x < 106	451	347	104	0.23059867	0.15101	0.07477	0.0870	-0.151013	0.00184057
106	106 ≤ x < 109	370	280	90	0.24324324	0.22097	0.06033	0.0753	-0.220967	0.00329661
109	109 ≤ x < 113	442	345	97	0.21945701	0.08711	0.07434	0.0811	-0.087114	0.00058943
113	113 ≤ x < 118	392	277	115	0.29336735	0.47686	0.05969	0.0962	-0.476862	0.0173904
118	118 ≤ x < 125	344	243	101	0.29360465	0.47801	0.05236	0.0844	-0.478006	0.01533863
125	x ≥ 125	384	268	116	0.30208333	0.51855	0.05775	0.0970	-0.51855	0.02034988
			4641	1196					0.8116173	0.18091413



BUREAU_SCORE	BINS	TOTAL	GOODS	BADS	BADS/TOTAL	WOE	GOOD%	BAD%	WOE	IV
-Infinity	x < 610	386	206	180	0.46632124	1.22103	0.04439	0.15050	1.1017098	-0.1169076
610	610 ≤ x < 632	389	231	158	0.40616967	0.97612	0.04977	0.13211	0.6482319	-0.053371
632	632 ≤ x < 647	394	266	128	0.3248731	0.62448	0.05732	0.10702	0.4642502	-0.023077
647	647 ≤ x < 659	397	266	131	0.32997481	0.64765	0.05732	0.10953	0.6181338	-0.0322768
659	659 ≤ x < 671	387	297	90	0.23255814	0.16202	0.06399	0.07525	0.1804656	-0.0020313
671	671 ≤ x < 681	381	302	79	0.20734908	0.01497	0.06507	0.06605	0.1093772	-0.0001073
681	681 ≤ x < 692	402	323	79	0.19651741	-0.0523	0.06960	0.06605	-0.26151	-0.0009267
692	692 ≤ x < 695	396	274	122	0.30808081	0.54684	0.05904	0.10201	0.0916477	-0.0039379
695	695 ≤ x < 706	384	323	61	0.15885417	-0.3108	0.06960	0.05100	-0.2081757	-0.0038708
706	706 ≤ x < 717	391	338	53	0.13554987	-0.4968	0.07283	0.04431	-0.151013	-0.0043061
717	717 ≤ x < 729	379	345	34	0.08970976	-0.9612	0.07434	0.02843	-0.220967	-0.0101445
729	729 ≤ x < 745	395	366	29	0.07341772	-1.1794	0.07886	0.02425	-0.087114	-0.0047577
745	745 ≤ x < 762	391	362	29	0.0741688	-1.1684	0.07800	0.02425	-0.476862	-0.0256327
762	762 ≤ x < 782	388	374	14	0.03608247	-1.9293	0.08059	0.01171	-0.478006	-0.0329253
782	x ≥ 782	377	368	9	0.02387268	-2.3549	0.07929	0.00753	-0.51855	-0.0372154
			4641	1196					0.8116173	-0.3514881



WOE helps establish a clear, monotonic relationship between predictor variables and the target, improving model performance. Monotonic binning is essential in credit risk modelling because it

improves model stability, interpretability, and predictive accuracy. It ensures that the relationship between a feature and risk is logical and consistent, helping lenders make better-informed credit decisions.

3.INFORMATION VALUE:-

Information Value (IV) quantifies the predictive power of an independent variable in distinguishing between good (non-default) and bad (default) cases. It is calculated using the difference in good and bad proportions across bins, weighted by the Weight of Evidence (WOE). Higher IV values indicate stronger predictive capability.

- Feature Selection – Helps identify the most relevant variables for credit scoring models.
- Predictive Power Assessment – Determines how well a variable differentiates between low- and high-risk borrowers.
- Model Stability – Ensures variables contribute meaningfully, reducing overfitting.
- Improved Risk Classification – Enhances accuracy in predicting defaults.
- Regulatory Compliance – Aids in developing transparent, explainable credit risk models.

To calculate Information Value (IV):

- Bin the Variable – Divide the independent variable into discrete bins.
- Calculate IV Contribution for Each Bin:

$$IV_{bin} = \big(P(Good) - P(Bad)\big) \times WOE$$

- Sum IV Values Across All Bins:

$$IV = \sum IV_{bin}$$

BUREAU_SCORE	BINS	TOTAL	GOODS	BADS	BADS/TOTAL	WOE	GOOD%	BAD%	WOE	IV
-Infinity	x < 610	386	206	180	0.46632124	1.22103	0.04439	0.15050	1.1017098	-0.1169076
610	610 ≤ x < 632	389	231	158	0.40616967	0.97612	0.04977	0.13211	0.6482319	-0.053371
632	632 ≤ x < 647	394	266	128	0.3248731	0.62448	0.05732	0.10702	0.4642502	-0.023077
647	647 ≤ x < 659	397	266	131	0.32997481	0.64765	0.05732	0.10953	0.6181338	-0.0322768
659	659 ≤ x < 671	387	297	90	0.23255814	0.16202	0.06399	0.07525	0.1804656	-0.0020313
671	671 ≤ x < 681	381	302	79	0.20734908	0.01497	0.06507	0.06605	0.1093772	-0.0001073
681	681 ≤ x < 692	402	323	79	0.19651741	-0.0523	0.06960	0.06605	-0.26151	-0.0009267
692	692 ≤ x < 695	396	274	122	0.30808081	0.54684	0.05904	0.10201	0.0916477	-0.0039379
695	695 ≤ x < 706	384	323	61	0.15885417	-0.3108	0.06960	0.05100	-0.2081757	-0.0038708
706	706 ≤ x < 717	391	338	53	0.13554987	-0.4968	0.07283	0.04431	-0.151013	-0.0043061
717	717 ≤ x < 729	379	345	34	0.08970976	-0.9612	0.07434	0.02843	-0.220967	-0.0101445
729	729 ≤ x < 745	395	366	29	0.07341772	-1.1794	0.07886	0.02425	-0.087114	-0.0047577
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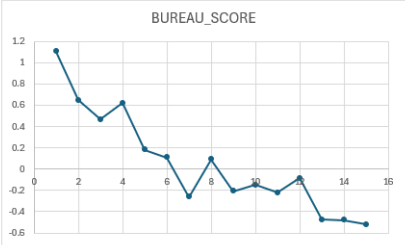
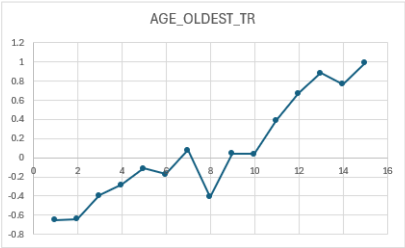
782	x ≥ 782	377	368	9	0.02387268	-2.3549	0.07929	0.00753	-0.51855	-0.0372154
			4641	1196					0.8116173	-0.3514881



AGE_OLDEST_TR	BINS	TOTAL	GOODS	BADS	BADS/TOTAL	WOE	GOODS%	BAD%	WOE	IV
-Infinity	x < 28	396	265	131	0.33080808	0.65141	0.05710	0.10953	-0.651415	0.03415498
28	28 ≤ x < 51	381	256	125	0.32808399	0.63908	0.05516	0.10452	-0.639083	0.03154166
51	51 ≤ x < 68	398	288	110	0.27638191	0.39347	0.06206	0.09197	-0.393467	0.01177161
68	68 ≤ x < 88	393	293	100	0.25445293	0.28094	0.06313	0.08361	-0.280945	0.00575349
88	88 ≤ x < 109	394	306	88	0.22335025	0.1097	0.06593	0.07358	-0.109699	0.0008386
109	109 ≤ x < 124	384	294	90	0.234375	0.17218	0.06335	0.07525	-0.172177	0.00204932
124	124 ≤ x < 136	395	319	76	0.19240506	-0.0785	0.06874	0.06355	0.0785106	0.00040747
136	136 ≤ x < 142	386	278	108	0.27979275	0.41046	0.05990	0.09030	-0.410457	0.01247795
142	142 ≤ x < 156	390	313	77	0.1974359	-0.0465	0.06744	0.06438	0.0464506	0.00014219
156	156 ≤ x < 173	388	311	77	0.19845361	-0.04	0.06701	0.06438	0.0400403	0.00010531
173	173 ≤ x < 196	378	322	56	0.14814815	-0.3933	0.06938	0.04682	0.3932527	0.00887133
196	196 ≤ x < 227	388	343	45	0.11597938	-0.6751	0.07391	0.03763	0.6751208	0.0244941
227	227 ≤ x < 268	384	347	37	0.09635417	-0.8825	0.07477	0.03094	0.8824597	0.0386799
268	268 ≤ x < 327	393	351	42	0.10687023	-0.7672	0.07563	0.03512	0.7671694	0.03108048
327	x ≥ 327	389	355	34	0.0874036	-0.9898	0.07649	0.02843	0.9898101	0.04757427
			4641	1196					1.2155705	0.24994268



IV helps assess the predictive power of a variable in credit risk modelling and feature selection.

Results	Interpretation
< 0.02	Insignificant predictor
0.02 - 0.1	Weak predictor
0.1 - 0.3	Medium predictor
0.3 - 0.5	Strong predictor
> 0.5	Highly predictive (potential overfitting risk)

2.Important Points

- Information value increases as bins / groups increases for an independent variable. When there are more than 20 bins as some bins may have a very few numbers of events and non-events.
- Information value is not an optimal feature (variable) selection method when you are building a classification model other than binary logistic regression (for eg. random forest or SVM) as conditional log odds (which we predict in a logistic regression model) is highly related to the calculation of weight of evidence. In other words, it's designed mainly for binary logistic regression model. Also think this way - Random Forest can detect non-linear relationship very well so selecting variables via Information Value and using them in random forest model might not produce the most accurate and robust predictive model.

3.Conclusion on WOE and IV:-

- WOE and IV play a vital role in building reliable credit risk models by refining data and selecting strong predictors. WOE transforms variables into meaningful values, ensuring

consistency in risk assessment, while IV quantifies their importance in distinguishing good and bad borrowers. Their combined use enhances model performance, enabling better decision-making in credit scoring and risk management. By leveraging these techniques, financial institutions can develop more accurate, transparent, and data-driven lending strategies.